Day 4: Component-Specific Troubleshooting Practical (1 hour): Identify and fix issues in faulty components (RAM, GPU, storage). Task: Create a checklist for diagnosing hardware issues.

1.Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

ANS:

Step 1: Turn off the computer and unplug the power cable.

Step 2: Open the computer case or laptop cover carefully.

Step 3: Find the RAM module on the motherboard.

Step 4: Open the RAM slot clips and remove the RAM carefully.

Step 5: Check the RAM for dust or damage.

Step 6: Put the RAM back into the slot in the correct direction and press gently until it locks into place.

Step 7: Close the computer case and connect the power cable.

Step 8: Turn on the computer and check whether it starts properly.

**2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.**

ANS:

1. Check the display: See if the screen is blank or showing lines.
2. Check the cable: Make sure the monitor cable is connected properly.
3. Check the GPU: Turn off the PC and check whether the graphics card is fitted properly.
4. Check the power cable: Make sure the GPU power cable is connected properly.
5. Check the driver: Update the graphics card driver.
6. Check the temperature: Make sure the GPU is not too hot.
7. Test the game: Run a game and check if it works properly.
8. Confirm the fix: Restart the PC and check whether the problem is solved.

3.Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

ANS:

Steps

1. Turn off the PC: Switch off the computer and unplug the power cable.

2. Find the drive: Open the computer case and find the SSD or HDD.

3. Disconnect the drive: Carefully unplug the drive's cables.

4. Start the PC: Turn on the computer and check the BIOS. The drive may not appear, or you may see “No Bootable Device Found.”

5. Turn off the PC again: Unplug the power cable.

6. Reconnect the drive: Connect the cables properly. For an NVMe drive, make sure it is fitted correctly.

7. Check the BIOS: Turn on the PC and see whether the drive is detected.

8. Check the result: If the drive appears again, restart the PC and check whether it starts normally.

4.Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

ANS:

1. Which component will I check first?

I will check the RAM first because faulty or loose RAM can make the computer restart again and again.

2. Steps to Check RAM

1. Turn off the PC: Switch off the computer and unplug the power cable.
2. Open the PC case: Remove the side cover.
3. Check the RAM: See if the RAM is loose.
4. Remove the RAM: Carefully remove it from the slot.
5. Reinsert the RAM: Put it back properly.

6. Start the PC: Turn on the computer and check if it works.
7. Check the result: See if the PC still restarts.

3. Check Other Components

- GPU: Check the graphics card and its cables.
- Storage: Check the SSD or HDD connections.